

AI CRM Lead Automation

A safe, testable workflow that converts inbound messages into structured CRM records, priorities, follow-up dates, and spreadsheet-safe exports.

DELIVERY	ANALYSIS	QUALITY	DEPLOYMENT
Local dashboard + API	Offline or optional OpenAI	50 automated tests	Local-first application

The business problem

Small teams receive inquiries through forms, inboxes, chat, and referrals. Manual interpretation and copy-paste work delay responses, lose context, and make it hard to see which lead needs attention first.

The implemented solution

The local application validates each intake, extracts contact details, classifies the request, calculates a transparent Hot/Warm/Cold score, schedules follow-up, checks duplicate emails, and writes an audit event. A responsive dashboard, JSON endpoints, and CSV export make the result immediately reviewable.

Two analysis modes, one contract

DEFAULT: DETERMINISTIC	OPTIONAL: OPENAI
Offline, repeatable, zero API cost. Versioned scoring policy feeds transparent priority and follow-up logic.	PII-redacted input, strict Structured Outputs, store: false, bounded retries, and local output redaction.

Workflow

1 Bounded JSON intake Media type, byte count, object shape, source, and message length are validated.	2 Contact + classification Contacts are extracted locally; provider input and generated fields are redacted.	3 Atomic CRM write Lead, duplicate state, and event log commit in one versioned atomic state replacement.
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Built for credible review

An inspectable, production-oriented architecture with explicit privacy, security, and operating controls.

Reliable persistence	Safer interfaces
- Transactional lead + event state - Atomic tempfile + replace commit - Schema-version and corruption checks - Legacy JSON migration	- Generic server errors - HTML escaping and allowlisted classes - Formula-safe CSV cells - No-cache and browser security headers

Privacy by default

- Raw inquiry text is not retained unless STORE_RAW_MESSAGE=1 is deliberately enabled.
- API keys, provider bodies, and lead text are excluded from logs.
- Contact PII is removed locally before OpenAI; generated stored fields are redacted again.
- Responses application-state storage is disabled with store: false.

Verification evidence

50 isolated tests	3 Python versions in CI	2 rebuilt + validated PDFs
Analysis, extraction, provider, storage, concurrency, CSV, HTML, and local HTTP behavior.	Python 3.11, 3.12, and 3.13 run the same standard-library suite.	Build, page count, metadata, and required text are checked in CI.

Client adaptation path

After requirements discovery, the same validated record can be connected to an authenticated form or inbox and written through an official CRM API. Production delivery includes identity and access controls, TLS, rate limits, a managed transactional database, retention/deletion automation, monitoring, backups, and reconciliation.

Integration scope

The architecture provides documented integration points for Gmail, HubSpot, Airtable, Google Sheets, Pipedrive, GoHighLevel, Zapier, Make, and n8n. Connector selection, credentials, and data mapping are configured for each client environment. OpenAI connectivity is enabled with the project owner's API key; all provider-independent behavior is locally and continuously verified.